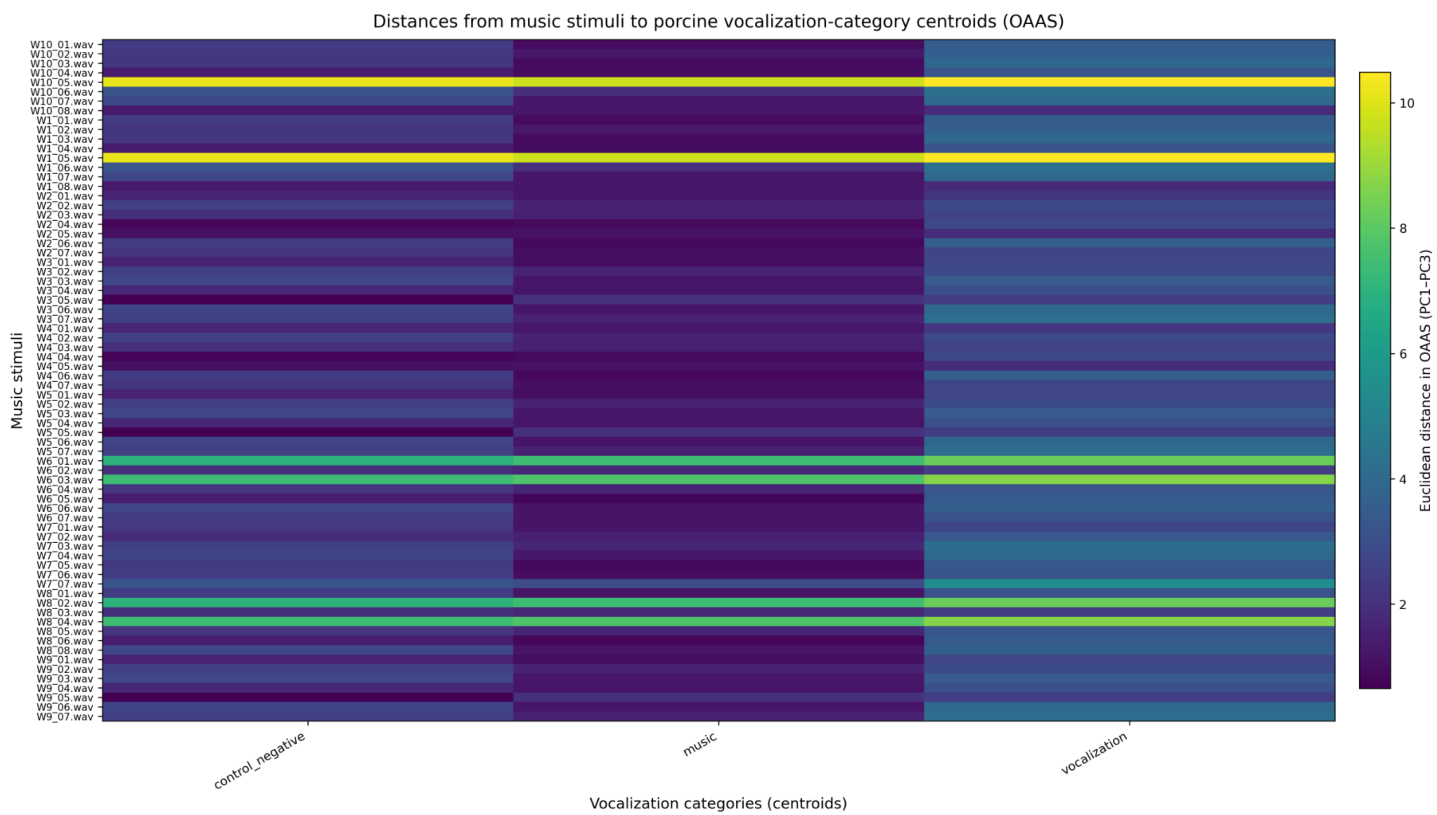
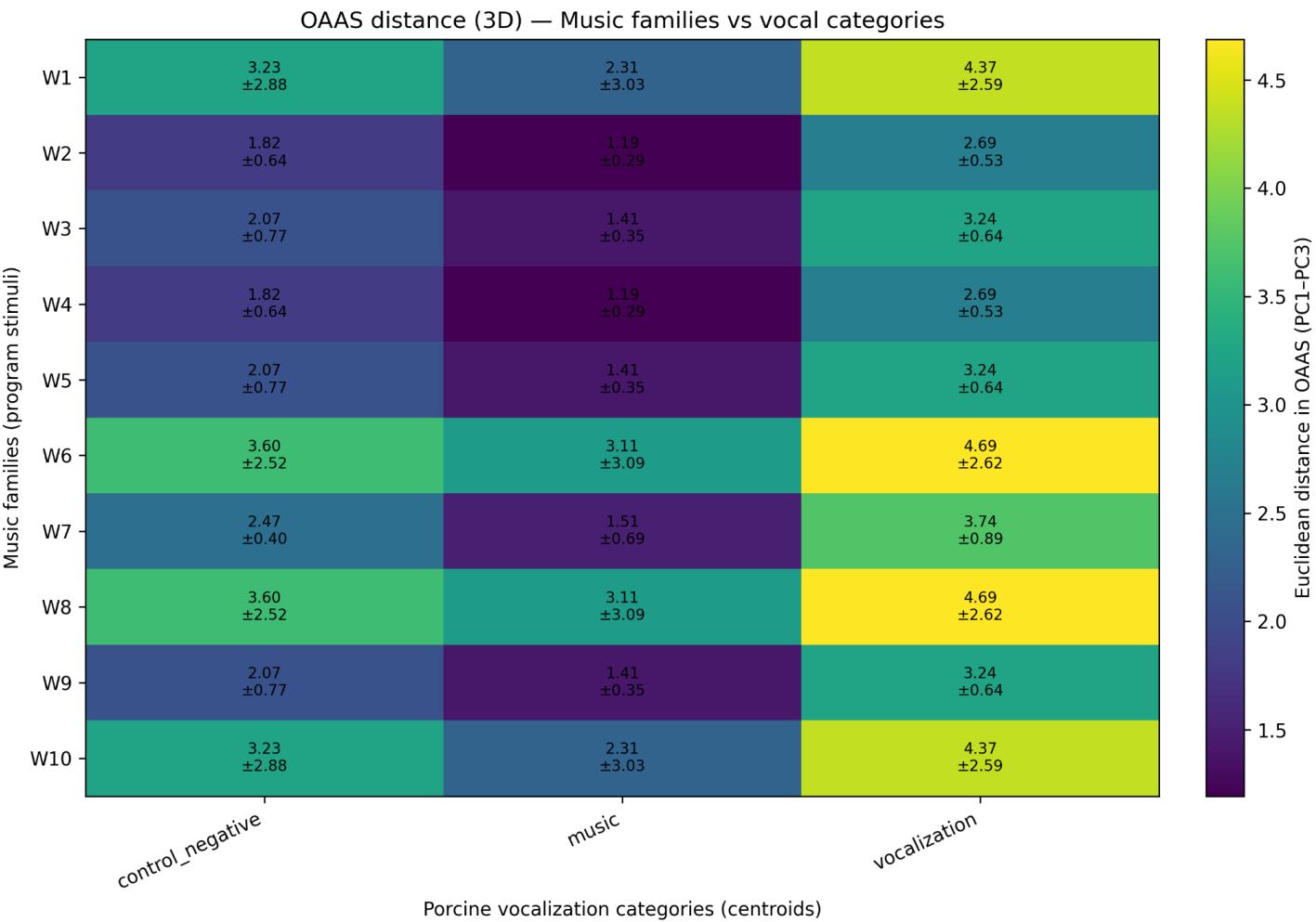


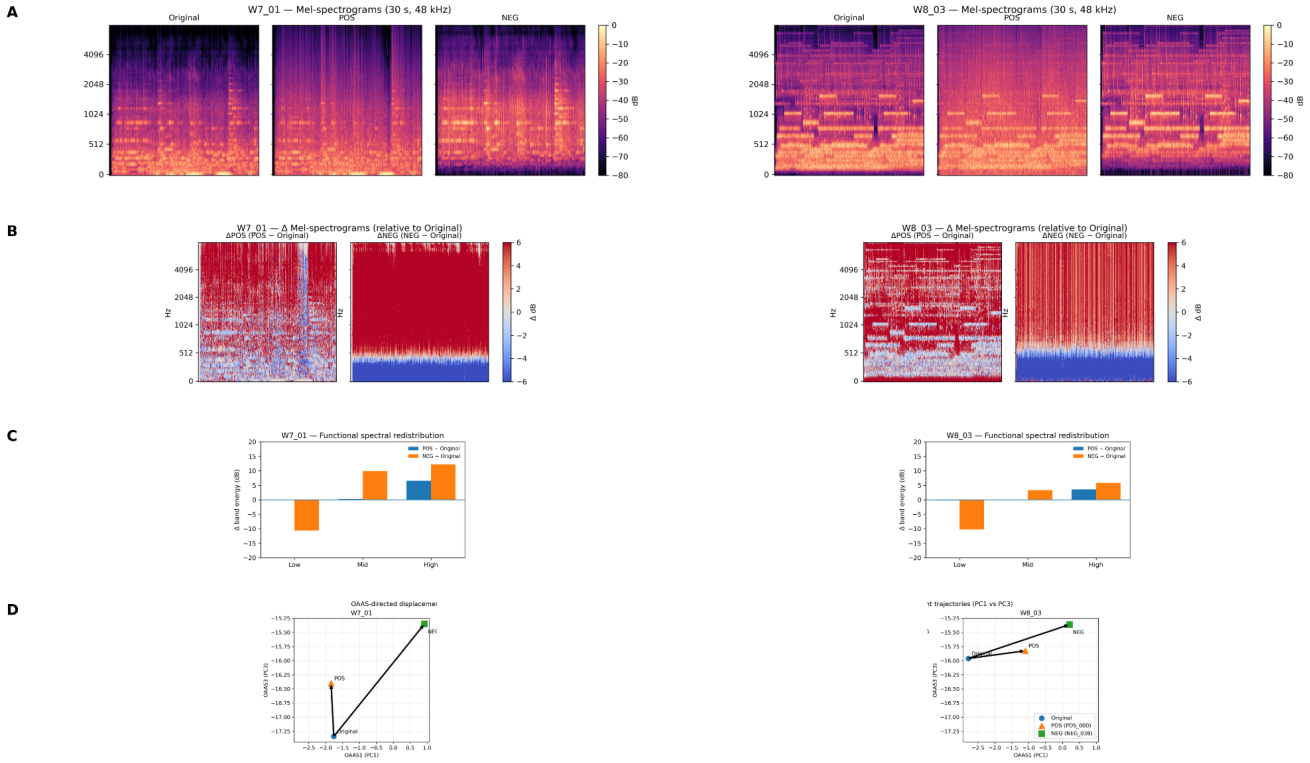
Supplementary Figure S1. Three-dimensional Euclidean distances (PC1–PC3) between individual music stimuli and porcine vocalization-category centroids in OAAS. The heatmap preserves full 3D geometry and resolves proximity patterns across all stimuli.



Supplementary Figure S2. Mean $\pm$ SD three-dimensional Euclidean distances (PC1–PC3) between music families (W1–W10) and porcine vocalization-category centroids in OAAS. This aggregated view summarizes family-level proximity patterns and complements Supplementary Figure S1.



Supplementary Figure S3. Comparative OAAS-directed transformations of W7_01 and W8_03.



Multilevel visualization of OAAS-guided positive (POS) and negative (NEG) transformations for stimuli W7_01 and W8_03. Panels show (A) mel-spectrograms (30 s, 48 kHz), (B) Δ mel-spectrograms relative to the original, (C) band-wise spectral redistribution (Δ energy across low, mid, and high frequency bands), and (D) OAAS displacement trajectories in the joint embedding (PC1–PC3 plane).